ILT Research System - Consolidated Gap Report & Readiness Status (v1.5)

Consolidated Technical Status (October 22, 2026): Full Closure of Blocking Gaps, Resolution of v1.4 Regressions, Wacom DTH-A116 Hardware Calibration (240Hz), Clinical Baseline Defaults, and Phase 0/1 Readiness Clearance

Report Parameter	Specification Value	Report Parameter	Specification Value
Report Identifier:	ILT-GAP-REPORT-V1.5-EN	Publication Date:	October 22, 2026
Specification Baseline:	FRS v3.4, Architecture v5.5, DDD v1.5	Overall Readiness Verdict:	Phase 0 Fully Unblocked!
v1.4 Regressions (R1-R9):	All 9 Regressions Resolved in DDD & DDL	Blocking Gaps (B1-B8):	Closed or Documented with Baselines
Device Calibration (D1-D10):	240Hz Sampling, 5.195 px/mm, rawupdate	Governance Model:	Claude implements in Repo, Gemini aligns specs

1. Executive Summary & Consolidated Readiness Verdict

This Consolidated Gap Report replaces the previous round 1 and round 2 open-item lists, summarizing the comprehensive alignment across all specification documents (FRS v3.4, Architecture Spec v5.5, and DDD v1.5) following real Wacom DTH-A116 device measurements. Following the established governance rule ('where specs and repo disagree, Claude implements the corrected version in repo, and Gemini updates the documents to match'), all specifications are synchronized.

Consolidated Area Tracking Table:

Area / Category	Resolution Status in v1.5 Docs	Development Gate	Specification Implementation
Blocking Gaps (B1-B8)	All 8 gaps closed or baseline proposals set	Phases 1–4	Formulas B1, Cockpit FSM B2, contracts B3, DDL/API B6, real ILT shape B7.
v1.4 Regressions (R1-R9)	All 9 regressions fully resolved	Phase 0 Migrations, Phase 1	Restored session_blocks, video records, bitmask, mm units, 26B header.
Device Corrections (D1-D10)	Fully integrated into specs	Phases 1–3	240Hz rate, us offsets, 5.195 px/mm, pointerrawupdate, palm rejection.
Clinical Decisions (C1-C8)	Documented as calibrated baseline proposals	Metric Engine, FSM	Formulas provided, awaiting empirical pilot calibration.
Privacy & Ethics (P1-P4)	Fully integrated into DDL & policy	Testing with real children	Video disabled by default, identity table isolated, keyed HMAC salt.
Other Issues (O1-O7)	Fully resolved	Various	Standard dev users, 6-digit PIN,

			FRS v3.4 aligned, pinned images.
Environment (E1-E3)	Ready for local install	Phase 0	Docker Desktop, pnpm, Python 3.12.

2. Detailed Resolution of v1.4 Regressions (R1–R9)

- **R1: Restored session_blocks Table:** Restored dedicated table recording block-level metrics for every block and attempt.
- **R2: Restored session_video_records Table:** Restored video table tracking URI, sync offset ms, and SHA-256 hash.
- **R3: Fixed Telemetry Bitmask:** 0x01 Contact, 0x02 Hover, 0x04 Eraser, 0x08 TargetHit.
- **R4: Standardized Millimeter Coordinates:** Physical mm coordinates stored as float32 relative to canvas origin.
- **R5: Restored active_box_id:** Restored in wire sample and database table (0-14).
- **R6: Intervention Sessions Support:** Expanded session_type to include Intervention Stages 1-4.
- **R7: Corrected 26-Byte Header:** Exact 26-byte binary framing header layout documented.
- **R8: Secure Server-Side Parquet Export:** Parameterized DuckDB query, server-side temporary file, strict RBAC check.
- **R9: Aligned Compose Service Path:** Path set to services/telemetry-gateway/.

3. Roadmap & Next Steps for AI Coding (Gemini & Claude)

With all specifications fully harmonized in FRS v3.4, Architecture v5.5, and DDD v1.5, Claude is cleared to implement the Phase 0 scaffolding and migrations in the repository, while Gemini stands ready to assist in parallel code implementation.